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Doping Engineering Enables Highly Conductive and Thermally Stable N-type Organic Thermoelectrics with High Power Factor

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KEYWORDS: doping engineering, electrical conductivity, power factor, thermal stability, fullerene derivative

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3 ABSTRACT: This work exploits the scope of doping engineering as an enabler for better-
4 performing and thermally stable n-type organic thermoelectrics. A fullerene derivative with polar
5 triethylene glycol type side chain (PTEG-1) is doped either by “co-processing doping” with n-
6 type dopants such as n-DMBI and TBAF or by “sequential doping” through thermal deposition
7 of Cs_2CO_3 . Solid-state diffusion of Cs_2CO_3 appears to dope PTEG-1 in the strongest
8 manner, leading to the highest electrical conductivity of $\sim 7.5 \text{ S/cm}$ and power factor of 32
9 $\mu\text{W}/(\text{mK}^2)$. Moreover, the behavior of differently doped PTEG-1 films under thermal stress is
10 examined by electric and spectroscopic means. Cs_2CO_3 -doped films are most stable, likely due to
11 a coordinating interaction between the polar side chain and Cs^+ based species, which
12 immobilizes the dopant. The high power factor and good thermal stability of Cs_2CO_3 -doped
13 PTEG-1 make it very promising for tangible thermoelectric applications.
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1. INTRODUCTION

Recently, the prospect of realizing lightweight, economical, and mechanically flexible electricity generation modules, has led to increased activity in research concerning organic thermoelectric (TE) materials.¹⁻⁵ The performance of a TE material is expressed by a dimensionless figure of merit, $ZT = S^2\sigma T/\kappa$. Where S , σ , κ , and T are the Seebeck coefficient, electrical conductivity, thermal conductivity, and absolute temperature, respectively.⁴ In an organic material, molecules interact with each other through weak Van Der Waals force, enabling an poorly thermally conductive system with low thermal conductivity of <1 W/(mK). As such, their TE performances are determined by a parameter known as the power factor (PF):^{6,7} $PF = S^2\sigma$. Tangible TE applications require both p- and n-type conducting materials with high power factors and sufficient working stability under a thermal stress. However, the latter is still much less developed than the former.⁸⁻¹⁵ In this regard, growing research efforts turn to design better n-type TE systems.

Aiming at optimizing the power factor of an n-type TE system, a tunable number of free charges is enabled by doping through electron transfer from n-dopant to host material. Adachi and coworkers reported an evaporated C₆₀ / cesium carbonate (Cs₂CO₃) bilayer, which offers a σ of ~ 8 S/cm and a $S^2\sigma$ of ~ 20.5 μ W/(mK²).¹⁶ However, with a few exceptions,^{8,17-21} most n-doped solution-processed organic systems exhibit a σ of <1 S/cm and a $S^2\sigma$ of <10 μ W/(mK²).²²⁻²⁴ For instance, Chabinye and co-workers reported a σ of $\sim 10^{-3}$ S/cm and a $S^2\sigma$ of ~ 0.1 μ W/(mK²) for a doped Naphthalenediimide (NDI)-based copolymer by a 1H-benzoimidazole derivative (n-DMBI).¹¹ Afterwards, considerable effort has been directed to enhance the n-doping efficiency of NDI-based copolymers by improving the host/dopant miscibility,²⁵⁻²⁸ planarizing the backbone,^{23,29} and tailoring the

density of states.²⁴ Particularly, the polar side chains have been used by us and other researchers to improve the system miscibility and thus the electrical conductivity.^{20,27,30–32} This led to a σ of 0.3 S/cm and $S^2\sigma$ of 0.4 $\mu\text{W}/(\text{mK}^2)$ for conjugated polymers and ~ 2 S/cm and 16.7 $\mu\text{W}/(\text{mK}^2)$ for fullerene derivatives, respectively.^{20,33} Alternatively, Yee and coworkers reported an n-type nickel ethenetetrathiolate (NiETT) polymer based composite, which shows a σ of >1 S/cm and a $S^2\sigma$ of >10 $\mu\text{W}/(\text{mK}^2)$.³⁴ Most of previous works focused on developing better host materials while how to choose an n-type dopant and the doping method are rarely explored. Moreover, it is unknown whether or not those previously reported n-doped organic systems can withstand a thermal stress in an operating model. Very recently, Kemerink and coworkers reported an “inverse-sequential” doping method for a fullerene derivative, which offers approximately 6 times higher σ and $S^2\sigma$ compared to the traditional “co-processing” (directly alloying the host and dopant in solid).³⁵ This study undercores the significance of the doping method for n-type organic thermoelectrics.

Here, we explored the impact of different doping methods not only on the performance but also on the thermal stability of n-type organic thermoelectrics. This is a case study on a fullerene derivative with a polar triethylene glycol-type side chain (PTEG-1). PTEG-1 was doped either by blending with an n-type dopant (n-DMBI or TBAF) or by solid-state diffusion of inorganic salts like Cs_2CO_3 . Cs_2CO_3 was found to be the most effective dopant for PTEG-1, giving the highest σ of ~ 7.5 S/cm. Doping PTEG-1 with TBAF and n-DMBI gave the optimized σ of 3.2 S/cm and 2 S/cm, respectively. Finally, PTEG-1 films doped by n-DMBI, TBAF and Cs_2CO_3 showed optimized $S^2\sigma$ of 16 $\mu\text{W}/(\text{mK}^2)$, 24

$\mu\text{W}/(\text{mK}^2)$ and $32 \mu\text{W}/(\text{mK}^2)$, respectively. These values put them among the best performing n-type organic thermoelectrics. Additionally, the different doping methods led to a variation in film density, which appears to dictate the thermal stability of the doped films. Doping with Cs_2CO_3 yielded the most stable films. Our study offers an insight into the effect of doping methods on the thermoelectric performance and the thermal stability of doped films.

2. RESULTS AND DISCUSSION

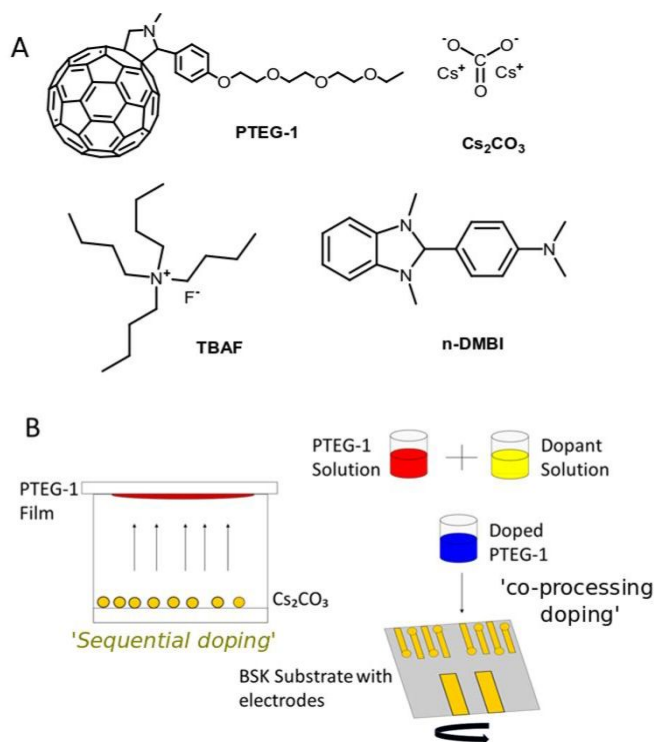


Figure 1. (A) The chemical structures of the host (PTEG-1) and the dopants (TBAF, n-DMBI and Cs_2CO_3); (B) the schematic of the doping processes (“sequential doping” and “co-processing doping”).

2.1 Doping Engineering and Electrical Conductivity.

Figure 1 shows the chemical structures of host and dopants, as well as the corresponding schematic doping processes. Previous studies indicated that PTEG-1 can be effectively doped by n-DMBI.^{20,36} Although Cs_2CO_3 and TABF have proven effective dopants for fullerene-based materials such as C_{60} and PCBM,^{16,37} they have not been used to dope PTEG-1 and the corresponding effects on the relevant n-type thermoelectrics are yet to be reported. Similar to the doping process with n-DMBI, TBAF was mixed with PTEG-1 in chloroform and the mixture was used to prepare the doped PTEG-1 film in an one-step process. This process is known as “co-processing doping”. Alternatively, PTEG-1 was doped by the inorganic salt Cs_2CO_3 in a two-step process, in which the inorganic salt was thermally deposited on top of solution-cast neat PTEG-1 film. We refer to this process as “sequential doping”. Earlier studies on Cs_2CO_3 indicated that Cs_2CO_3 decomposes into cesium oxide during thermal evaporation.³⁸ In the present study, the Cs_2CO_3 decomposition was evidenced by gasing during the evaporation process. Therefore, the real dopant may be cesium oxide instead of Cs_2CO_3 . However, as this work is not focused on the doping mechanism, for simplicity, we still refer to this as “ Cs_2CO_3 doping”. It is speculated that Cs_2CO_3 (or a resulting species) diffuses into the bulk of the host film and enables the n-doping process.

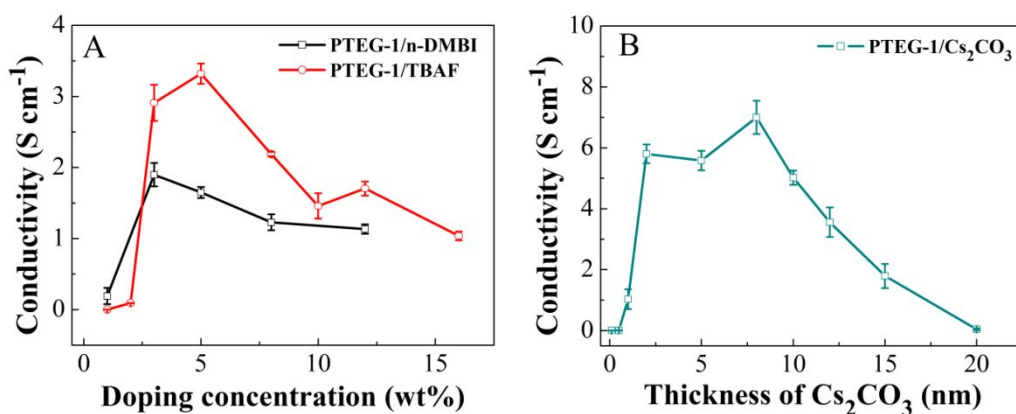


Figure 2. N-doping PTEG-1 with different dopants. (A) plots of electrical conductivity versus doping concentration in the doped PTEG-1 by n-DMBI and TBAF; (B) electrical conductivity as a function of thickness of Cs_2CO_3 layer.

Figure 2 displays the electrical conductivity of doped PTEG-1 films after three different doping processes, and measured by a standard four-point probe method (see the details in the experimental section). The neat PTEG-1 film exhibited a very low σ of 10^{-9} S/cm. As shown in Figure 2a, the electrical conductivity of the PTEG-1 films greatly increased upon blending with n-DMBI and TBAF. The doped PTEG-1 films exhibited a peak σ of 2 S/cm and 3.2 S/cm at a doping concentration of 3 wt% for n-DMBI and 5 wt% for TBAF, respectively. By further increasing the dopant loading, the electrical conductivity dropped in both cases, likely because the high concentration of dopant molecules disrupted the film microstructure.¹⁸ Figure 2b shows the electrical conductivity of PTEG-1 films doped by “sequential doping” with evaporated Cs_2CO_3 . It should be pointed out that it is very difficult to determine how much of the deposited Cs_2CO_3 really diffuses into the active layer to n-dope the material. However, by controlling the Cs_2CO_3 thickness, it is possible to qualitatively modulate the amount of Cs_2CO_3 diffusing into the active layer. By increasing the thickness of the Cs_2CO_3 layer, the electrical conductivity increased to a peak at 8 nm and then decreased. A best σ of ~ 7.5 S/cm was achieved, representing the highest value for n-doped fullerene derivatives. One may have a concern regarding whether ion movement contributes to the charge transport in doped films with salts such as TBAF and Cs_2CO_3 . To have an insight on this point, we carried out several sets of experiments to examine the possible ionic behavior and the results are displayed in Figure S1 in the Supporting Information (SI). The results indicate that for the doped PTEG-1 films with salts, ions hardly move within the bias range used in this study and capacitive

behavior was not apparently seen. As such, we believe that the ionic transport is negligible in our doped organic systems.

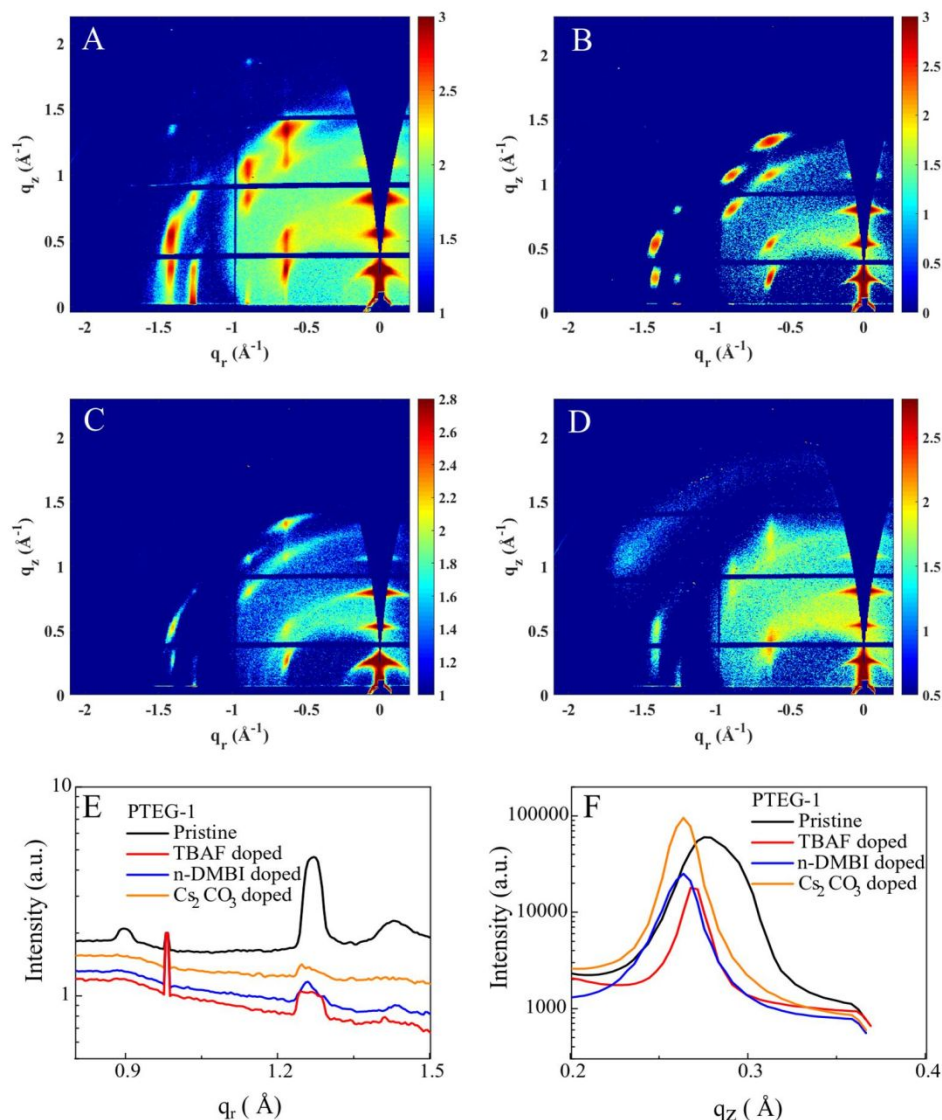


Figure 3. 2D GIWAXS patterns of neat PTEG-1 (A) and doped PTEG-1 films with TBAF (B), n-DMB (C) and Cs_2CO_3 (D); and their corresponding linecuts (E) in the in-plane direction and (F) in the out-of-plane direction.

2.2 Molecular Packing.

The microstructure of various PTEG-1-based films was investigated by two-dimensional (2D) grazing incidence wide angle X-ray scattering (GIWAXS). The corresponding 2D-

GIWAXS patterns and linecuts are shown in **Figure 3**. Four strong scattering peaks are clearly seen along the near out-of-plane q_z direction for neat PTEG-1 film (Figure 3a). These signals are attributed to the (00 l) family of reflections, suggesting that PTEG-1 packs into a layered structure parallel to the substrate. The layer planes contain the buckyball and the glycolated side chain. The (001) peak position for the neat PTEG-1 thin film determines a spacing of 2.25 nm. The π - π stacking peak of fullerene cages along the in-plane direction is at $q_r=1.23 \text{ \AA}^{-1}$, which corresponds to a spacing of 0.51 nm. These results are in line with our previous report.²⁰ The doping processes appear to not change the orientation of PTEG-1 molecules and π -stacking spacing (Figures 3b-3f). However, the q_z (001) peak shifts to larger spacings, depending on the doping process. For doped PTEG-1 films by co-processing with n-DMBI and TBAF, the (001) spacing is shifted to 2.40 nm, while for “sequential doping” with Cs_2CO_3 it shifted to 2.33 nm. Based on these observations, we assume that the dopants are mainly located in the plane of the polar side chains, and the different (001) spacings are caused by the varying sizes of the dopants. However, it is unclear whether or not the dopants were also incorporated in the amorphous regions. The crystallinity of PTEG-1 film was reduced more upon Cs_2CO_3 doping than in the case of “co-processing doping”. The reason behind is still unknown, and we surmise that it is correlated with the polar side chain and the unique molecular stacking of PTEG-1.

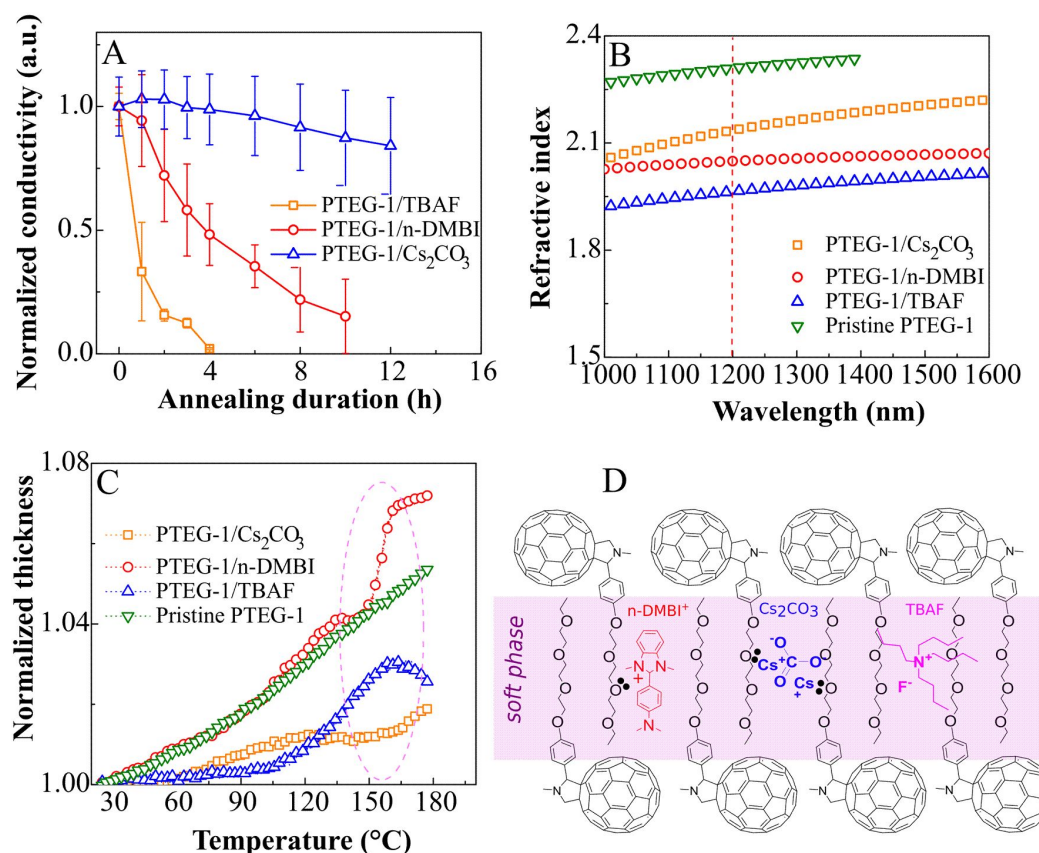


Figure 4. Thermal stability of PTEG-1-based films doped by various dopants. (A) Degradation of electrical conductivities in doped PTEG-1 films with three dopants (TBAF, n-DMBI and Cs₂CO₃) upon thermal stress (150 °C); (B) refractive index n spectra of various PTEG-1-based thin films; (C) In situ (normalized) thickness evolution of various PTEG-1-based films with growing temperature detected in dynamic ellipsometry model; (D) the schematic of the interaction between the dopants and host (the doping mechanism is not fully demonstrated).

2.3 Electric and Spectroscopic Characterization of Stability under a thermal stress.

The operating stability of doped organic films under a thermal stress is very critical to practical thermoelectric applications. As such, the dopant should be carefully selected, both with regard to of the thermoelectric performance and thermal stability of the TE film. **Figure 4a** displays the evolution of the electrical conductivity of doped PTEG-1 films upon heating at 150 °C for different durations in inert atmosphere. The PTEG-1 film doped with Cs₂CO₃ (“sequential doping”) proved most stable and maintained 80% of its

initial electrical conductivity after thermal annealing for 12 h. In contrast, the PTEG-1 films doped by “co-processing doping” showed a much faster decrease in electrical conductivity. The TBAF-doped PTEG-1 film completely lost its electrical conductivity after 4 h while the n-DMBI-doped PTEG-1 film degraded to 5% of its initial conductivity. Previous studies attributed such a loss of electrical conductivity to a dedoping process, which is related to the diffusion of dopant molecules out of the host matrix upon the thermal annealing.^{20,39} In this sense, an improved thermal stability may indicate less thermal annealing-driven spacial mobility of the dopant molecules.

Incorporating extrinsic dopants into a host matrix is expected to modify the molecular packing and thus the density (ρ) and polarizability of the film. The variations in film density and polarizability reflect into changes in the refractive index (n).^{40,41} As the molecular orientation of PTEG-1 is not affected by the doping process (as indicated by GIWAXS data), the refractive index is mainly a function of the film density. In the region of the spectrum where PTEG-1 does not absorb, the variation in refractive index can be expressed as:⁴⁰

$$\frac{n_t^2 - 1}{n_0^2 - 1} \cong \frac{\rho_t}{\rho_0} \quad (1)$$

Clearly, a larger refractive index corresponds to a higher density of the film. The UV-vis-NIR absorption spectra of various PTEG-1-based films were measured to determine the transparency in the region of 1000-1700 nm (**Figure S2** in the SI). We measured the refractive index (n) of various PTEG-1-based films in such a spectral region by non-invasive variable angle ellipsometry (See Figure 4b). The neat PTEG-1 film displays the highest refractive index $n=2.3$ (at 1200 nm) and is therefore assigned as being the most dense film. Upon doping, n decreases, indicating that the films become less dense. The order of the density of the PTEG-1 films is neat > Cs₂CO₃-doped > n-DMBI-doped >

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3 TBAF-doped. The de-doping process upon thermal annealing is considered to be related
4 to the diffusion of dopant and the resultant demixing of the host and dopant.^{31,39} If the
5 doped PTEG-1 film is more closely packed, there is less space for dopant molecules to
6 diffuse, which may lead to a better thermal stability. In this way, the density of doped
7 PTEG-1 film explains the variation of thermal stability shown in Figure 4a.

14 Variable temperature ellipsometry is a powerful tool to detect the real-time phase
15 behavior of organic films via examining the thickness variation with temperature.^{40,41}
16 Figure 4c displays the evolution of thickness of the PTEG-1-based films with temperature
17 (25°C-180°C), measured by variable temperature ellipsometry in inert atmosphere. At
18 temperatures above 150 °C, the doped PTEG-1 films by TBAF and n-DMBI showed a
19 fast change in thickness with elevating temperature, attributing to the demixing of the
20 host and dopant molecules. In contrast, the PTEG-1 film doped with Cs₂CO₃ displayed a
21 plateau with the slowest change in the thickness. These results explain the thermal
22 stability characteristics of PTEG-1 films doped by various dopants, which is related to the
23 demixing of the host and dopant molecules driven by the thermal stress. As mentioned
24 above, the (cationic) dopants are expected to be mainly incorporated in the plane of the
25 polar side chains. This allows us to discuss the microstructural origin of the different
26 behaviors of the doped films under thermal stress. It has been well established that the
27 ethylene oxide-based organic species can chelate alkali salts by forming a complex with
28 the cationic species through interaction with the lone pairs of the oxygen atoms of the
29 ether linkages.^{42,43} This parallels the famous and quite strong crown ether-cation
30 interactions. We postulate that the triethylene glycol type side chain may, to some extent,
31 stabilize Cs₂CO₃ (or resulting cationic dopant species) and thus make it less mobile under
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thermal stress as illustrated in Figure 4d. Hence, the Cs_2CO_3 doped PTEG-1 film exhibits the best thermal stability in this study. As for the case of TBAF doping, the counter-ion TBA^+ possesses the most bulky structure and therefore, is expected to have the weakest interaction with the surroundings composed by the glycolated side chains. As such, the PTEG-1-based film doped by TBAF renders the worst stability under a thermal stress.

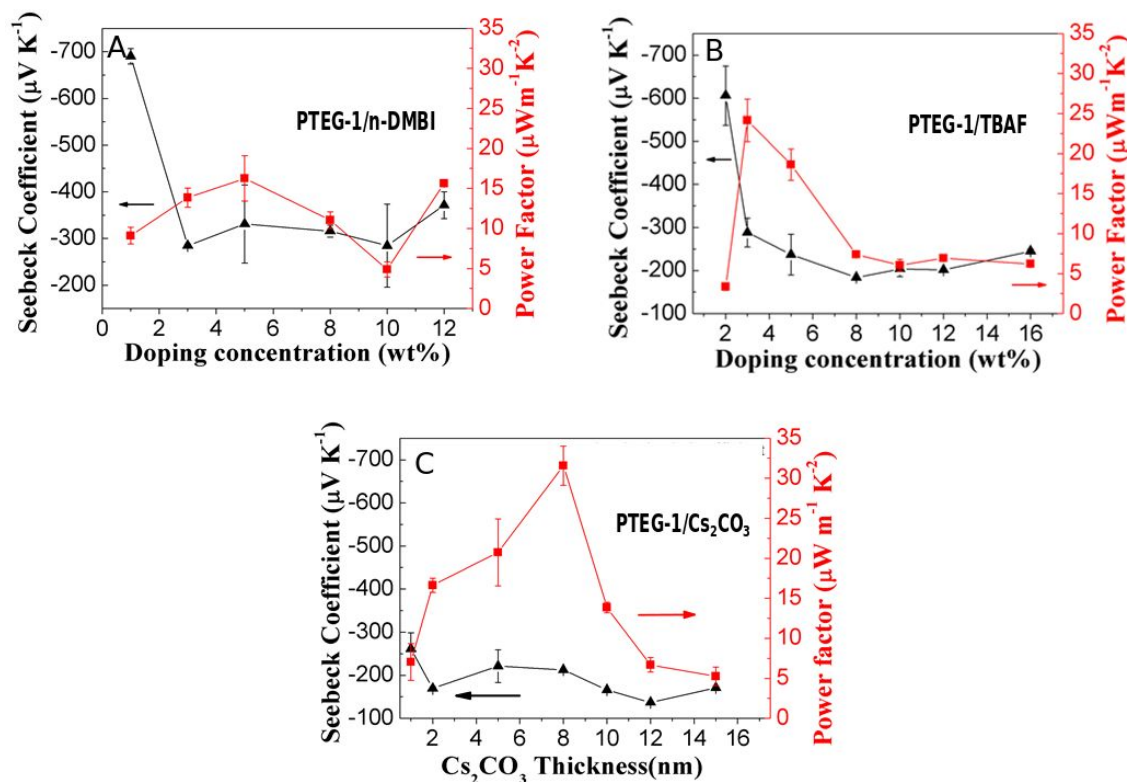


Figure 5. Thermoelectric parameters (S and $S^2\sigma$) plots for PTEG-1-based films doped by n-DMBI (A), TBAF (B) and Cs_2CO_3 (C).

2.4 Thermoelectric Performance

The energy offset between the Fermi level energy (E_F) and the charge transport energy (E_T) governs the Seebeck coefficient.^{24,44} By furthering the doping, more charge carriers are produced, causing a shift of E_F towards E_T and a decrease in the absolute S value. As such, S can be used to indirectly evaluate the doping level. We measured S values of PTEG-1 films doped with the three different dopants and calculated the power factors

accordingly, which are displayed in **Figures 5a-5c**. All of the doped PTEG-1 films show negative S , indicating the electron as the dominating mobile charge carrier. The PTEG-1 film doped with 1 wt% n-DMBI exhibited a S of $-690 \mu\text{V/K}$. By furthering the n-DMBI loading, the S rapidly decreased in absolute value to $-330 \mu\text{V/K}$ at 5 wt% and finally reached a plateau with more dopant. Similarly, the PTEG-1 film doped with TBAF showed a S of $-606 \mu\text{V/K}$ at 2 wt%, which increased to $-288 \mu\text{V/K}$ at 3 wt% and finally reached a regime with little change by further doping. The optimized $S^2\sigma$ of $16 \mu\text{W}/(\text{mK}^2)$ and $24 \mu\text{W}/(\text{mK}^2)$ were achieved at doping concentrations of 5 wt% and 3 wt% for n-DMBI and TBAF, respectively. Figure 5c shows the S and $S^2\sigma$ of PTEG-1 film doped with Cs_2CO_3 . In this scenario, the doping level of the PTEG-1 film can be modulated by the thickness of deposited Cs_2CO_3 layer. When the thickness of Cs_2CO_3 was 1 nm, the S was $-286 \mu\text{V/K}$, which decreased in absolute value to $-212 \mu\text{V/K}$ with 8 nm-thick Cs_2CO_3 layer. The smaller S of PTEG-1 films doped with Cs_2CO_3 than those obtained in the case of n-DMBI and TBAF doping indicates the higher doping level for the former. This explains the higher σ for PTEG-1-based film doped by Cs_2CO_3 . Accordingly, the highest $S^2\sigma$ of $32 \mu\text{W}/(\text{mK}^2)$ was obtained, which is among the best n-type organic thermoelectrics (see **Table 1**).

Table 1. The list of thermoelectric parameters of solution-based n-type OTEs.

Host	Dopant (method)	Conductivity [S/cm]	Seebeck coefficient [$\mu\text{V/K}$]	Power factor [$\mu\text{W}/(\text{mK}^2)$]	Ref.
PTEG-1	Cs_2CO_3 (evaporate)	7.5	-212	32	this work
PTEG-1	n-DMBI (solution)	1.5	-330	16	this work

PTEG-1	TBAF (solution)	2.9	-288	24	this work
C₆₀	Cs ₂ CO ₃ (evaporate)	8	-160	20	Ref.16
PTeEG-1	n-DMBI (solution)	2.3	-319	23	Ref.18
A-DCV-DPPTT	n-DMBI (solution)	3.1	-568	105	Ref.8
TEG-N2200	n-DMBI (solution)	0.17	-153	0.4	Ref.27
p(gNDI-gT2)	n-DMBI (solution)	0.3	-93	0.4	Ref.28
FBDPPV	n-DMBI (solution)	14	-140	28	Ref.21
PNDTI-BBT-DP	n-DMBI (solution)	5	-169	14	Ref.17
PNDI2TEG-2Tz	n-DMBI (solution)	1.8	-159	4.6	Ref.24
P(NDI2OD-Tz2)	TDAE (vapor)	0.06	-447	1.5	Ref.23
PDPF	n-DMBI (solution)	1.3	-235	4.7	Ref.19

3. CONCLUSIONS

To summarize, we studied the impact of different doping methods on the n-type thermoelectric properties of organic films. PTEG-1, with its characteristic polar side chain is used as the host, which is doped either by a one-step process with n-DMBI and TBAF as the dopant or by a two-step process with solid-state diffusion of Cs₂CO₃. Doping with Cs₂CO₃ gives the highest σ of ~ 7.5 S/cm with the best $S^2\sigma$ of $32 \mu\text{W}/(\text{mK}^2)$ in this study, outperforming both the TBAF and n-DMBI-based systems. The PTEG-1 molecules stack into a bilayered architecture parallel to the substrate. Cs₂CO₃ doping appears to exert the smallest influence on the spacing of the side chain plane likely due to its relatively small size, compared with the other two. In a similar sense, PTEG-1 film doped by Cs₂CO₃ shows the highest density. Owing to these advantages, Cs₂CO₃-doped PTEG-1 film exhibits the best stability under a thermal stress (150 °C). The high power

factor and good thermal stability of the PTEG-1/Cs₂CO₃ doping system make it very promising for tangible thermoelectric applications.

4. EXPERIMENTAL SECTION

4.1 Materials. PTEG-1 was synthesized by our group. TBAF, n-DMBI and Cs₂CO₃ were purchased from Sigma Aldrich.

4.2 Thermoelectric Characterization. Clean borosilicate glass substrates were treated with UV-ozone for 20 minutes. The doped films were spin-coated from a fullerene derivative solution (10 mg/mL in chloroform) blended with different amounts of dopant solution (20 mg/mL in chloroform) in a N₂-filled glovebox. The resultant films were subjected to thermal annealing at 120 °C for 1 hour. The electrical conductivity measurements by four-point probes were carried out in an N₂-controlled environment. The electrical conductivity was calculated with $\sigma = (I/V) \times L / (w \times d)$, with L (1 mm); w (4.5 mm); d (100 nm), the length, width and height of the channel, respectively. The electrical conductivities of the separate points were averaged to obtain the electrical conductivity of one device. The Seebeck coefficients of doped PTEG-1 thin-film samples were measured in a home-built setup, as reported previously.¹⁹ Two Peltier devices are placed with a gap of 6 mm in parallel on a heat sink. The temperatures of two Peltier devices can be controlled by two independent PID controllers (proportional-Integral-Derivative Controllers). The two rectangular Au electrodes (width: 1.5 mm and length: 6 mm) are deposited on glass substrate with a distance of 6 mm. The large distance is used for avoiding the contact geometric effect. Two T-type thermocouples (127 μ m from Omega) are used as probes for simultaneously temperature and thermal voltage. Note that a silver paste (ELECTRODAG 1415) was used to

connect thermocouple probes with the Au electrodes. A Keithley 2000 mounted with scanner card is used for signal recording with a delay time of 100 ms. The system is controlled by Labview software. In order to remove the thermal voltage (ΔV) shifting, A 'quasi-static' approach by slowly changing temperature difference (ΔT) is used to extract the Seebeck coefficient.

4.3 Microstructure Characterization. Grazing incidence wide-angle X-ray scattering (GIWAXS) measurements were performed using a MINA X-ray scattering instrument built on a Cu rotating anode source ($\lambda=1.5413 \text{ \AA}$) by following the precedures reported previously.^{45,46}

4.4 Optical constant and phase behavior. The various PTEG-1-based films were prepared on clean silicon substrates with a thin layer of native oxide. The optical constants were measured by variable angle ellipsometry (J. A. Wollam Co., inc). The cauchy dispersion function was used for fitting to ellipsometry data to obtain refractive index in the transparent region. For the phase behavior measurement, the samples were placed in an air-protected sample holder filled with N_2 , where variable temperature spectroscopic scan was carried out (the incident angle of 70°). The film thickness variation with temperature was followed using the previously reported procedure.³⁶ Phase change leads to a change in volumetric expansion coefficient, which is directly translated into the tangent slope variation of ρ - T function and hence the d - T function.

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Notes

The authors declare no competing financial interest.

Author Contributions

J. L. conceived this work. J. L. wrote the manuscript. M. P. G. and J. L. conducted the device fabrication and characterizations. J. D. and G. P. conducted microstructural characterization. B. Z. conducted ellipsometry test. L. Q. and J. C. H. provided the host material. J. L. and L. J. A. K. supervised this project. The manuscript was examined by all authors. The final manuscript was approved by all the coauthors.

ASSOCIATED CONTENT

The Supporting Information Available:[Ionic transport characterization, Figure S1; Absorption spectra. Figure S2]

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FIGURE CAPTIONS

Figure 1. (A) The chemical structures of the host (PTEG-1) and the dopants (TBAF, n-DMBI and Cs_2CO_3); (B) the schematic of the doping processes (“sequential doping” and “co-processing doping”).

Figure 2. N-doping PTEG-1 with different dopants. (A) plots of electrical conductivity versus doping concentration in the doped PTEG-1 by n-DMBI and TBAF; (B) electrical conductivity as a function of thickness of Cs_2CO_3 layer.

Figure 3. 2D GIWAXS patterns of neat PTEG-1 (A) and doped PTEG-1 films with TBAF (B), n-DMB (C) and Cs_2CO_3 (D); and their corresponding linecuts (E) in the in-plane direction and (F) in the out-of-plane direction.

Figure 4. Thermal stability of PTEG-1-based films doped by various dopants. (A) Degradation of electrical conductivities in doped PTEG-1 films with three dopants (TBAF, n-DMBI and Cs_2CO_3) upon thermal stress (150 °C); (B) refractive index n spectra of various PTEG-1-based thin films; (C) In situ (normalized) thickness evolution of various PTEG-1-based films with growing temperature detected in dynamic ellipsometry model; (D) the schematic of the interaction between the dopants and host (the doping mechanism is not fully demonstrated).

Figure 5. Thermoelectric parameters (S and $S^2\sigma$) plots for PTEG-1-based films doped by n-DMBI (A), TBAF (B) and Cs_2CO_3 (C).

TOC GRAPHICS

